Forecasting with Machine Learning: Forecast Evaluation and Comparison

Forecasting with Machine Learning

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Outline for Today

- **Designing an Out-of-Sample Forecasting Exercise**
- **Estimation and Forecasting Schemes**
- **Forecast Evaluation Across Applications**
- **Statistical Comparison of Forecasts**
- **Economic Value and Reliable Forecasting Research**

Designing an Out-of-Sample Forecasting Exercise

Pseudo-Out-of-Sample Forecasting

IDEA

Use historical data (e.g., 2000–2025) to simulate forecasts that could have been made in real time.

FORECAST

At each forecast origin t, estimate or update the model using available information and predict

$$\widehat{Y}_{t+1|t} = \widehat{f}_t(X_t).$$

EVALUATE

Once Y_{t+1} is observed, record the forecast error

$$e_{t+1|t} = Y_{t+1} - \widehat{Y}_{t+1|t}.$$

Repeat over the test period to obtain out-of-sample forecast errors.

Key Principle: Each forecast uses only information available at its forecast origin.

Training, Validation, and Testing

- In ML forecasting, the data typically serve three distinct purposes.

Sample	Example	Purpose
Training	2000–2015	Estimate model parameters
Validation	2016–2020	Select models and tune hyperparameters
Testing	2021–2025	Evaluate the selected forecasting procedure

- Example: select the LASSO penalty λ based on forecasting performance during 2016–2020.
- The test period should not influence the choice of λ or other model decisions.
- After model selection, the chosen specification may be re-estimated using all pre-test data.

Why Chronological Splitting Matters

- Random train-test splits may use future observations for training (e.g., 2024 data to predict 2018 outcomes).
- They may also ignore time dependence and changes in economic relationships.
- Chronological splitting preserves the time ordering and tests whether a model generalizes to genuinely future observations.

Information Availability and Look-Ahead Bias

- Chronological splitting alone does not prevent **information leakage (look-ahead bias)**. Examples:

Publication delays Using 2025 earnings in January 2026, before the annual report is released.

Data revisions Using today's revised GDP figures to simulate forecasts made in 2020.

Preprocessing Standardizing 2020 data using means calculated with data through 2025.

Model selection Choosing the best model after examining test-period performance.

> **Practical Principle**: Prevent avoidable leakage. Account for publication delays and data revisions where feasible, and acknowledge remaining limitations.

Estimation and Forecasting Schemes

Updating Forecasting Models Over Time

- As new observations become available, we must decide whether and how to update the forecasting model.
- Three common approaches:

1	**Fixed estimation**	Estimate the model once and keep its parameters unchanged.
2	**Expanding window**	Re-estimate using all available historical observations.
3	**Rolling window**	Re-estimate using a fixed number of the most recent observations.

Three Forecasting Schemes: Formal Definition

- Let $\hat{\theta}_t$ denote the model parameters used at forecast origin t.
- Let D_s denote an observation available for estimation at date s, and let T_0 be the initial estimation endpoint.

FIXED ESTIMATION

$$\hat{\theta}_t = \hat{\theta}(D_1, \ldots, D_{T_0}).$$

EXPANDING WINDOW

$$\hat{\theta}_t = \hat{\theta}(D_1, \ldots, D_t).$$

ROLLING WINDOW OF LENGTH W

$$\hat{\theta}_t = \hat{\theta}(D_{t-W+1}, \ldots, D_t).$$

- In each case, the one-step-ahead forecast is constructed as

$$\widehat{Y}_{t+1|t} = f(X_t; \hat{\theta}_t).$$

Illustration: Which Observations Are Used?

- Suppose we initially estimate a model using observations 1–5.
- We generate forecasts for periods 6, 7, and 8.

Scheme	Forecast 6	Forecast 7	Forecast 8
Fixed	1:5	1:5	1:5
Expanding	1:5	1:6	1:7
Rolling ($W = 5$)	1:5	2:6	3:7

- The table shows which observations are used to estimate each forecast.

Which Estimation Scheme Should We Use?

Scheme	Main advantage	Main limitation
Fixed	Simple; computationally cheap	Does not adapt to new data
Expanding	Larger estimation sample	Older data may be less relevant
Rolling	Adapts to changing relationships	Smaller sample; window choice

Key Trade-off: Estimation precision versus adaptation to structural change.

Forecasting Schemes in Machine Learning

- In ML forecasting, distinguish three decisions:

PREDICTION FREQUENCY

How often are forecasts generated?

RE-ESTIMATION FREQUENCY

How often are model parameters updated?

TUNING FREQUENCY

How often are hyperparameters reconsidered?

- These frequencies need not coincide. Consider this illustrative example from stock return forecasting:
  - Predict monthly.
  - Re-estimate annually.
  - Retune every three years.
- Periodic updating offers a compromise between fixed models and re-estimation at every forecast date, reducing computational costs, especially for complex ML models.

Forecast Evaluation Across Applications

Evaluating Forecasting Performance

- Suppose two models produce out-of-sample forecasts. How do we compare their performance?
- Four important considerations:

1	**Accuracy measure**	How are forecast errors measured?
2	**Benchmark**	What alternative should the model beat?
3	**Forecast horizon**	How far ahead are we predicting?
4	**Evaluation sample**	Which periods or observations are included?

- These choices can affect which forecasting method performs best.

Common Measures of Forecast Accuracy

- Given forecast errors $e_1, \ldots, e_P$, commonly used measures are:

$$\text{MSE} = \frac{1}{P}\sum_{t=1}^{P} e_t^2, \qquad \text{RMSE} = \sqrt{\frac{1}{P}\sum_{t=1}^{P} e_t^2}, \qquad \text{MAE} = \frac{1}{P}\sum_{t=1}^{P} |e_t|.$$

- **MSE/RMSE** penalize large errors more strongly than MAE.
- RMSE and MAE have the same units as the forecast target; MSE has squared units.
- The appropriate measure depends on the forecasting application.

Three Common Forecast Benchmarks

1. Historical mean

$$\widehat{Y}^{B}_{t+1|t} = \frac{1}{t}\sum_{s=1}^{t} Y_s.$$

Common benchmark for aggregate stock returns.

2. Random walk (no change)

$$\widehat{Y}^{B}_{t+1|t} = Y_t.$$

Common for persistent economic and financial series.

3. Autoregressive forecast

$$\widehat{Y}^{B}_{t+1|t} = \widehat{\alpha}_t + \hat{\rho}_t Y_t.$$

Exploits estimated historical dependence.

Important: Benchmarks must use the same information available to competing ML models.

Choosing Appropriate Forecast Benchmarks

- A forecasting method should be compared with a relevant benchmark.

Application	Typical benchmarks	Research examples
Stock returns	Zero return, historical mean, simple linear models	Gu et al. (2020)
Inflation	Autoregressive and other traditional macro forecasting models	Medeiros et al. (2021)
Volatility	HAR models using daily, weekly, and monthly volatility	Christensen et al. (2023)
Electricity prices	Seasonal naive forecasts, established autoregressive models	Lago et al. (2021)

Out-of-Sample R^2: The Benchmark Matters

- A general benchmark-relative measure is

$$R^2_{\text{OS}} = 1 - \frac{\sum_{t\in\mathcal{T}}(Y_t - \widehat{Y}_t)^2}{\sum_{t\in\mathcal{T}}(Y_t - \widehat{Y}_t^B)^2}.$$

- $R^2_{\text{OS}} > 0$ indicates better forecasting performance than the benchmark; $R^2_{\text{OS}} < 0$ indicates worse performance.
- The benchmark depends on the application:
  - Aggregate returns: Historical mean.
  - Individual stock returns (Gu et al., 2020): Zero excess return.
- In Gu et al. (2020), the sums run over all stock-month observations, giving an **uncentered panel** R^2.

Illustration: Interpreting Out-of-Sample R^2

- Suppose models A and B have out-of-sample mean squared errors:

$$\text{MSE}_A = 0.018, \qquad \text{MSE}_B = 0.020.$$

- Taking model B as the benchmark:

$$R^2_{\text{OS}} = 1 - \frac{0.018}{0.020} = 0.10.$$

- Model A reduces MSE by **10%** relative to model B.

> **Question**: Does a 10% reduction in MSE imply a statistically significant improvement?

Volatility Forecasting: A Different Evaluation Problem

- Volatility is not directly observed; realized variance RV_{t+1} serves as a proxy. For a variance forecast $\hat{v}_{t+1|t} > 0$, two common losses are:

$$L_{\text{MSE}} = (RV_{t+1} - \hat{v}_{t+1|t})^2,$$
$$L_{\text{QLIKE}} = \log(\hat{v}_{t+1|t}) + \frac{RV_{t+1}}{\hat{v}_{t+1|t}}.$$

- Both losses are robust to noise in a conditionally unbiased volatility proxy.
- Unlike MSE, QLIKE penalizes underprediction more strongly. Underestimating volatility may lead to excessive risk exposure.

> **Application**: Christensen et al. (2023) compare ML methods with HAR-type volatility forecasts.

Electricity Prices: Evaluating Hourly Forecasts

- Day-ahead forecasts predict 24 hourly prices. Let $p_{d,h}$ and $\hat{p}_{d,h}$ denote actual and forecast prices.
- For D days, the mean absolute error is

$$\mathrm{MAE} = \frac{1}{24D} \sum_{d=1}^{D} \sum_{h=1}^{24} |p_{d,h} - \hat{p}_{d,h}|.$$

- Lago et al. (2021) emphasize MAE and relative MAE (rMAE), strong benchmarks, and evaluation across markets and long test periods.
- Percentage errors (MAPE) can be misleading when prices are near zero.

What Can Average Forecast Accuracy Hide?

- Average forecast accuracy may hide important differences across applications:

Stock returns Small versus large firms?

Macroeconomics Recessions versus expansions?

Volatility Short versus long forecast horizons?

Electricity prices Different hours, seasons, or markets?

- Researchers therefore examine performance across subperiods, horizons, and groups of observations.

Statistical Comparison of Forecasts

Comparing Forecast Losses

- For two forecasting models A and B, define the **loss differential**:

$$d_t = L(Y_t, \widehat{Y}_t^A) - L(Y_t, \widehat{Y}_t^B).$$

- Under squared error loss:

$$d_t = (e_t^A)^2 - (e_t^B)^2.$$

  $d_t < 0$ favors model A; $d_t > 0$ favors B.

- The average loss differential is

$$\bar{d} = \frac{1}{P} \sum_{t=1}^{P} d_t.$$

- $\bar{d} < 0$ indicates lower average loss for A, but not necessarily a statistically significant improvement.

The Diebold–Mariano Test

- The **Diebold–Mariano (DM) test** compares the expected forecast losses of two models:

$$H_0 : \mathbb{E}[d_t] = 0.$$

- The test statistic is

$$DM = \frac{\bar{d}}{\widehat{\mathrm{se}}(\bar{d})}.$$

- Under suitable regularity conditions,

$$DM \xrightarrow{d} N(0,1) \qquad \text{under } H_0.$$

- At the 5% level:
  - **Two-sided**: Reject if $|DM| > 1.96$.
  - **One-sided**: Reject if $DM < -1.645$ when testing whether model A outperforms B.

Serial Dependence and the DM Test

- Forecast loss differentials d_t may be serially correlated, especially when forecast horizons overlap.
- Ignoring serial dependence can lead to incorrect standard errors and misleading significance tests.
- The DM statistic therefore uses a **HAC standard error** for $\bar{d}$, accounting for serial correlation in the loss differentials.

> **Caution**: Standard DM inference may require adjustments in certain settings, including nested models.

Illustration: Is the Improvement Significant?

- Recall our earlier example:

$$\text{MSE}_A = 0.018, \qquad \text{MSE}_B = 0.020.$$

- The average loss differential is

$$\bar{d} = 0.018 - 0.020 = -0.002.$$

- Suppose $\widehat{\text{se}}(\bar{d}) = 0.0008$:

$$DM = \frac{-0.002}{0.0008} = -2.5.$$

  The two-sided p-value is approximately 0.012: model A performs significantly better.

> **Key Distinction**: $R^2_{\text{OS}} = 10\%$ measures the improvement; DM tests its statistical significance.

Question: What if the standard error were twice as large?

Comparing Many Models: Model Confidence Set

- Suppose we compare five models: AR, LASSO, Random Forest, XGBoost, and Neural Network.
- Comparing five models requires 10 pairwise tests, creating a **multiple-comparison problem**.
- The **Model Confidence Set (MCS)** provides a joint procedure:

Test equal predictive ability. → If rejected, eliminate a poorly performing model. → Repeat until equality is no longer rejected.

- The remaining models form a **shortlist of best-performing candidates**, which cannot be statistically distinguished.

> **Illustration**: MCS = {Random Forest, XGBoost, Neural Network}. This does not imply identical true performance.

Economic Value and Reliable Forecasting Research

Forecast Accuracy versus Economic Value

- Statistical accuracy (MSE, MAE, OOS R^2) does not necessarily imply economic value.
- Example: Stock return prediction.
    - Model A predicts individual returns more accurately.
    - Model B better identifies which stocks will outperform.
    - Model B may be more useful for portfolio construction.

> **Main Distinction**: Statistical evaluation measures forecast accuracy; economic evaluation measures usefulness for particular decisions.

From Stock Return Forecasts to Portfolios

- An ML model predicts next month's return for each stock:

$$\hat{r}_{i,t+1|t} = \widehat{f}(X_{i,t}).$$

- Each month, rank stocks by predicted returns.
- Form two portfolios:
  - **High**: Highest predicted returns.
  - **Low**: Lowest predicted returns.
- A long-short strategy buys High and shorts Low:

$$R^{LS}_{t+1} = R^{High}_{t+1} - R^{Low}_{t+1}.$$

Application: Gu, Kelly, and Xiu (2020) evaluate ML forecasts using portfolios sorted by predicted returns.

Evaluating Portfolio Performance

- Key performance measures:

Average return How profitable is the strategy?

Sharpe ratio Return relative to volatility.

Risk exposures Exposure to market and other factors.

Turnover and trading costs Costs of portfolio rebalancing.

- For monthly long-short returns, the annualized Sharpe ratio is approximately

$$SR_{\text{ann}} \approx \sqrt{12}\,\frac{\overline{R^{LS}}}{s(R^{LS})}.$$

Important: Better return predictions need not imply higher net investment performance.

Searching Across Many Forecasting Models

- ML research often compares many specifications:
  - Predictors and transformations.
  - Algorithms and hyperparameters.
  - Sample periods and forecast horizons.
- Searching across many models can produce strong results by chance, exaggerating apparent forecasting gains.
- Model selection using **validation data** is legitimate.
- Repeatedly adjusting models after examining **test results** compromises out-of-sample evaluation.

> **Important Distinction**: MCS addresses comparisons among candidate models, but not repeated use of test data for model development.

What Makes Empirical Forecasting Evidence Convincing?

- Before accepting reported ML gains, consider five questions:

1	**Benchmarks**	Does ML beat strong alternatives?
2	**Evaluation design**	Was the test sample kept out of model development?
3	**Robustness**	Do results hold across periods, horizons, or markets?
4	**Economic relevance**	Are the gains useful after costs and risks?
5	**Transparency**	Are data and methods reproducible?

References

Christensen, K., Siggaard, M., & Veliyev, B. (2023). A machine learning approach to volatility forecasting. *Journal of Financial Econometrics*, 21(5), 1680–1727.

Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223–2273.

Lago, J., Marcjasz, G., De Schutter, B., & Weron, R. (2021). Forecasting day-ahead electricity prices: A review of state-of-the-art algorithms, best practices and an open-access benchmark. *Applied Energy*, 293, 116983.

Medeiros, M. C., Vasconcelos, G. F. R., Veiga, Á., & Zilberman, E. (2021). Forecasting inflation in a data-rich environment: The benefits of machine learning methods. *Journal of Business & Economic Statistics*, 39(1), 98–119.